PHYS 230 - Electricity and Magnetism
Lecture 3

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UNIVERSITY OF ALBERTA

Outline

1. Electric Charge and Electric Field
   - Coulomb's Law (Continued)
   - Fields
   - Electric Field

All figures in these slides are from one of the following sources, unless cited otherwise:

University Physics With Modern Physics, Young & Freedman

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Exercise 1

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Two equal positive charges $q_1 = q_2 = 2.0\ \mu$C are located at $x = 0$, $y = 0.30$ m and $x = 0$, $y = -0.30$ m, respectively. What are the magnitude and direction of the total electric force that q_1 and q_2 exert on a third charge $Q = 4.0\ \mu$C at $x = 0.40$ m, $y = 0$ m?

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Concept of a Field

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Scalar quantities filling all the space and time described by a **scalar field**.

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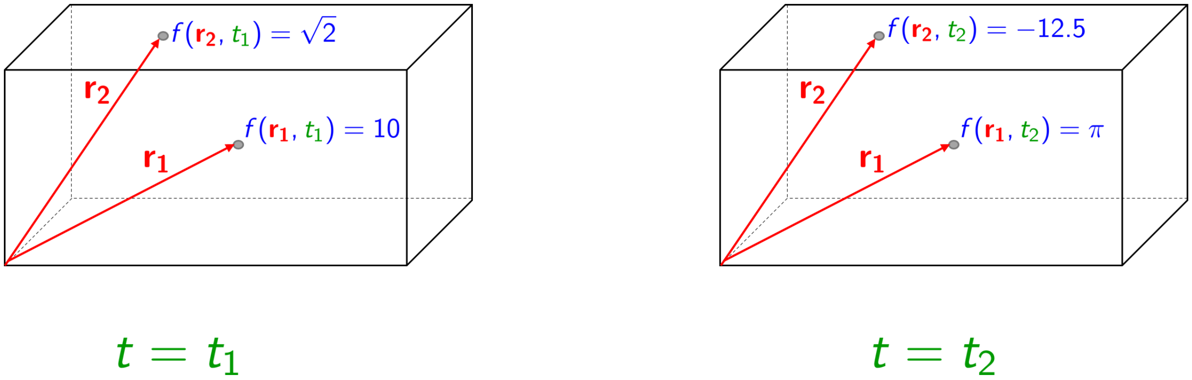

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Vector quantities filling all the space and time described by a **vector field**.

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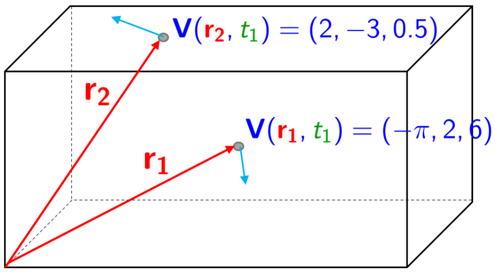
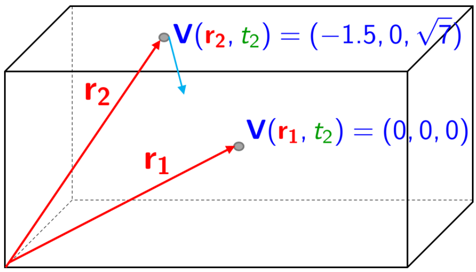


$t = t_1$



$t = t_2$

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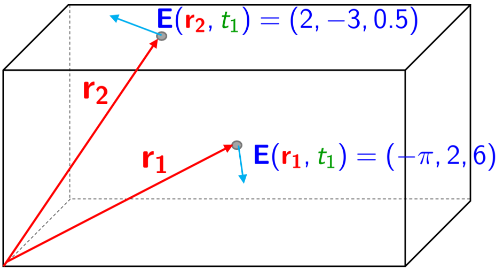

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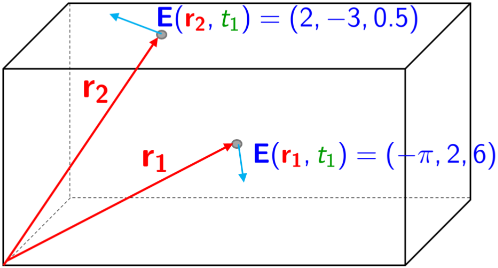

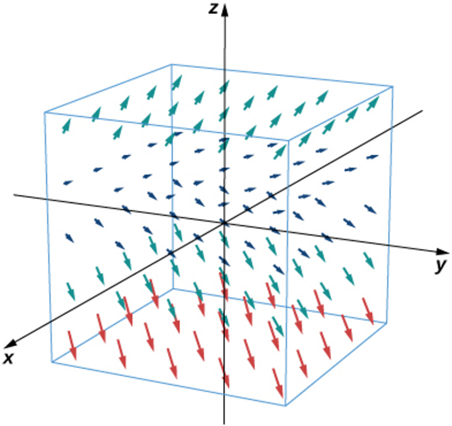

Electric Field and Electric Force

Any point charge q placed in an electric field $\mathbf{E}$ "feels" an electric force $\mathbf{F}$ on it, such that

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This is how electric field gives rise to electric force, i.e., the force on q resulting from the field $\mathbf{E}$ is

$$\mathbf{F} = q\mathbf{E}$$

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The SI units of electric field $\mathbf{E}$ is

$$[\mathbf{E}] = \frac{[\mathbf{F}]}{[q]} = \boxed{\frac{\mathrm{N}}{\mathrm{C}}}$$

Electric Field and Electric Force

The vector form of electric field due to a charge Q at the position of charge q can be computed as

$$\mathbf{E}_{Qq} = \frac{\mathbf{F}_{Qq}\overset{\frac{1}{4\pi\epsilon_0}\frac{Qq}{r_{Qq}^2}\hat{\mathbf{r}}_{Qq}}{\nearrow}}{q} = \frac{1}{4\pi\epsilon_0}\frac{Q}{r_{Qq}^2}\hat{\mathbf{r}}_{Qq}$$

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Form 3: Using full displacement vector from the source Q towards any point p (can be a point in empty space)

$$\mathbf{E}_{Qp} = \frac{1}{4\pi\epsilon_0}Q\frac{\mathbf{r}_p - \mathbf{r}_Q}{\left|\mathbf{r}_p - \mathbf{r}_Q\right|^3}$$

Direction of Electric Field

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So, intuitively:

To find the direction of field $\mathbf{E}_Q$, place a positive test charge q_0 in the field $\Rightarrow$ the direction of the force on q_0 due to $\mathbf{E}_Q$ is the direction of the field $\mathbf{E}_q$, since

$$\mathbf{F}_{q_0} = q_0\mathbf{E}_Q$$

Electric Field

Note: according to

$$\mathbf{E}_{Qp} = \frac{1}{4\pi\epsilon_0}\frac{Q}{r_{Qp}^2}\hat{\mathbf{r}}_{Qp}$$

the electric field at a point does not depend on the charge at that point! It only depends on the source

Principle of Superposition

The total electric field at a point p is the vector sum of the fields at p due to each point charge in the charge distribution

$$\mathbf{E}_p = \mathbf{E}_{1p} + \mathbf{E}_{2p} + \mathbf{E}_{3p} + \dots$$

Strategy To Find Electric Field Vector/Magnitude

1. Choose a coordinate system wisely
2. Depending on the problem, choose the formula (forms 1, 2, 3)
3. Divide computations into simpler steps, e.g., first find $\mathbf{r}_2 - \mathbf{r}_1$, then $|\mathbf{r}_2 - \mathbf{r}_1|$, etc.
4. Find **E** for each point charge; if necessary use the superposition principle to find the net force

Exercise 2

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What is the magnitude of the electric field **E** at a field point 2.0 m from a point charge $q = 4.0$ nC?

Exercise 3 https://epoll.srv.ualberta.ca – Code: MBD

A point charge $q = -8.0$ nC is located at the origin. Find the electric field vector at the point $\mathbf{r} = 1.2\hat{\mathbf{i}} - 1.6\hat{\mathbf{j}}$ m.

Exercise 3

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